

Critical Review and Reproduction Study on High-Dimensional Two-Sample Distribution Testing via Max-Sliced Wasserstein Distance

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Abstract

This report provides a critical review and our group’s reproduction study of the high-dimensional two-sample testing framework proposed by Hu and Lin (2025). High-dimensional nonparametric distribution testing faces significant difficulties due to the curse of dimensionality and signal dilution under sparse alternatives. The reviewed framework addresses these challenges by maximizing the one-dimensional 1-Wasserstein distance over a sparse subset of projection directions, coupled with a multiplier bootstrap procedure for null calibration. In this project, we implemented the entire computational pipeline from scratch and verified its size control and power advantages through controlled simulations. We also reproduced the real-world application on glioblastoma multiforme (GBM) and lower-grade glioma (LGG) DNA methylation data, successfully identifying global and marginal distributional discrepancies. In addition, we propose our own insights on the method’s core ideas, limitations and future directions, and provide derivations of several fundamental theoretical results.

1 Introduction and Related Work

The high-dimensional two-sample distribution test, which aims to determine whether two sets of high-dimensional samples are drawn from the same distribution, is a fundamental problem in nonparametric statistics and machine learning. Formally, given two independent samples $X_1, \dots, X_{n_1} \stackrel{\text{iid}}{\sim} P$ and $Y_1, \dots, Y_{n_2} \stackrel{\text{iid}}{\sim} Q$, where P and Q represent probability distributions on $\mathbb{R}^p$, the objective is to test the null hypothesis $H_0 : P = Q$ against the alternative hypothesis $H_1 : P \neq Q$. In contemporary biomedical studies, the dimensionality p frequently equals or exceeds the sample sizes n_1 and n_2 . Under these high-dimensional settings, the discrepancies between P and Q are often sparse, meaning they are confined to an unknown small subset of coordinates, which makes detection exceptionally challenging.

Early methods primarily include Maximum Mean Discrepancy (MMD) (Gretton et al., 2012) based on kernel embeddings, Energy Distance (ED) (Szekely and Rizzo, 2004) based on pairwise sample distances (Sejdinovic et al., 2013), as well as graph-based (Friedman and Rafsky, 1979) and nearest-neighbor-based test approaches. These methods exhibit favorable applicability in general distribution difference detection, but are prone to power loss as dimensionality increases, particularly under weak signals or sparse alternatives where differences are concentrated in only a handful of coordinates. (Ramdas et al., 2015; Gao and Shao, 2023; Zhu and Shao, 2020; Schilling, 1986) To provide a deeper structural analysis of these traditional routes, a comprehensive taxonomy detailing their specific statistical mechanisms and practical limitations is compiled in Table 4 in Appendix A.

In recent years, Wasserstein distance (WD) (Villani et al., 2009; Ramdas et al., 2017) has garnered significant attention for its ability to capture the geometric differences between

distributions, yet the naive empirical Wasserstein distance suffers from a severe curse of dimensionality in high-dimensional settings. (Lei, 2020) To this end, researchers have proposed a series of variants including sliced Wasserstein distance (SWD) (Bonneel et al., 2015), smoothed Wasserstein distance (Nietert et al., 2021), projected Wasserstein distance (PWD) (Wang et al., 2021a) and kernel projected Wasserstein distance (KRWD) (Wang et al., 2021b), which improve both statistical convergence properties and computational feasibility via smoothing or low-dimensional projection. The structural trade-offs and methodological limitations of these regularized variants are systematically evaluated in Table 5 in Appendix B.

The max-sliced Wasserstein distance (MSWD) research line pursued by (Hu and Lin, 2025; Deshpande et al., 2019) further emphasizes selecting from all possible projection directions the one that maximally amplifies the discrepancy between the two distributions. By integrating sparsity constraints with a multinomial multiplier bootstrap procedure, this framework successfully circumvents the dilution of localized signals, stabilizes the high-dimensional geometry, and bridges the gap between statistical power and computational feasibility.

2 Proposed Method: Sparse Max-Sliced Wasserstein Testing

The framework under review formalizes the structural discrepancy between P and Q using the s -sparse max-sliced Wasserstein distance. Let $\mathcal{V}_{0,s} = \{v \in \mathbb{R}^p : \|v\|_2 = 1, \|v\|_0 \leq s\}$ denote the set of all s -sparse unit vectors in $\mathbb{R}^p$, where s is a user-defined sparsity parameter. The population-level s -sparse max-sliced Wasserstein distance is defined as

$$D_{0,s}(P, Q) = \sup_{v \in \mathcal{V}_{0,s}} W_1(v^\top X, v^\top Y) = \sup_{v \in \mathcal{V}_{0,s}} \sup_{f \in \text{Lip}_1} \left\{ \mathbb{E}[f(v^\top X)] - \mathbb{E}[f(v^\top Y)] \right\}, \quad (1)$$

where Lip_1 represents the class of 1-Lipschitz functions mapping from $\mathbb{R}$ to $\mathbb{R}$. Given empirical samples, the plug-in sample estimator is constructed as

$$\hat{D}_{0,s}(P, Q) = \sup_{v \in \mathcal{V}_{0,s}} \sup_{f \in \text{Lip}_1} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} f(v^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(v^\top Y_j) \right\}. \quad (2)$$

A key observation is that 1-Wasserstein distance (W_1) is used here, rather than higher-order Wasserstein distances. Our group provides a dual-perspective insight into this choice. By the Kantorovich-Rubinstein theorem, the 1-Wasserstein distance enjoys a simple dual formulation defined on the space of 1-Lipschitz functions

$$W_1(P, Q) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left\{ \int_{\mathbb{R}^p} f dP - \int_{\mathbb{R}^p} f dQ \right\}. \quad (3)$$

This specific form allows the max-sliced statistic to be directly expressed as the supremum of an empirical process indexed by the class of 1-Lipschitz functions., making all standard tools of empirical process theory applicable. In contrast, the dual formulations of higher-order Wasserstein distances are complex and intractable, thus cannot be integrated into this empirical process framework. We leave further discussions on 1-Wasserstein distance in Appendix D.2.

To establish a pivotal test statistic, the empirical distance is scaled by a sample-size standardization factor. The global test statistic takes the form

$$T_n = \left(\frac{n_1 n_2}{n_1 + n_2} \right)^{1/2} \hat{D}_{0,s}(P, Q). \quad (4)$$

The scaling factor ensures that under the null hypothesis $H_0 : P = Q$, the empirical variance of the process remains stabilized, yielding a non-degenerate asymptotic distribution as sample sizes grow. (See Appendix D.3 for our analysis)

Since the supremum is taken over both an uncountably infinite function class Lip_1 and a non-convex combinatorial vector set $\mathcal{V}_{0,s}$, the null distribution of T_n is highly non-standard and cannot be analytically derived. The authors resolve this by employing a multinomial multiplier bootstrap procedure. Let $M = (M_1, \dots, M_{n_1})^\top \sim \text{Multinomial}(n_1; 1/n_1, \dots, 1/n_1)$ and $M' = (M'_1, \dots, M'_{n_2})^\top \sim \text{Multinomial}(n_2; 1/n_2, \dots, 1/n_2)$ be independent random vectors that represent bootstrap resampling weights. The centered multiplier bootstrap statistic is defined as

$$T_n^* = \left(\frac{n_1 n_2}{n_1 + n_2} \right)^{1/2} \sup_{v \in \mathcal{V}_{0,s}} \sup_{f \in \text{Lip}_1} \left\{ \frac{1}{n_1} \sum_{i=1}^{n_1} (M_i - 1) f(v^\top X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} (M'_j - 1) f(v^\top Y_j) \right\}. \quad (5)$$

By generating B independent bootstrap replicates $\{T_{n,b}^*\}_{b=1}^B$, the empirical $(1 - \alpha)$ -quantile $\hat{q}_{T^*}(1 - \alpha)$ is computed as the critical threshold. The global null hypothesis H_0 is rejected if the observed statistic T_n exceeds this threshold.

A major advantage of this multiplier bootstrap formulation is that it allows for the simultaneous construction of confidence intervals. For any fixed direction $v \in \mathcal{V}_{0,s}$, a simultaneous basic bootstrap confidence interval for the true projected distance $W_1(v^\top X, v^\top Y)$ can be established as

$$\text{SCI}\{W_1(v^\top X, v^\top Y)\} = \left[\hat{W}_1(v^\top X, v^\top Y) - \frac{\hat{q}_{T^*}(1 - \alpha)}{\sqrt{n_1 n_2 / (n_1 + n_2)}}, +\infty \right), \quad (6)$$

where $\hat{W}_1(v^\top X, v^\top Y)$ is the sorted one-dimensional empirical Wasserstein distance. Setting $v = e_k$, where e_k is the k -th canonical unit vector, provides an automatic family-wise error rate correction across all p marginal distributions without requiring any post-hoc adjustments.

3 Computational Implementation and Algorithmic Analysis

The practical feasibility of the sparse max-sliced Wasserstein test relies on evaluating the nested suprema efficiently. Our group analyzed the algorithmic logic, uncovering an elegant separation between the functional optimization over Lip_1 and the directional optimization over $\mathcal{V}_{0,s}$.

First, for any fixed projection direction v , the inner optimization over the infinite-dimensional function class Lip_1 drops out entirely due to the Kantorovich-Rubinstein theorem (Villani et al., 2009). In one dimension, the 1-Wasserstein distance has a well-known closed-form solution based on the horizontal integration of quantile functions. For two empirical univariate distributions, this distance can be computed exactly by sorting the projected observations. If two samples have equal sizes, the distance reduces to the mean absolute difference between the sorted coordinates. When sample sizes differ or bootstrap weights are applied, the evaluation is performed by matching cumulative step functions along the real line, which requires only $O(n \log n)$ operations.

Second, optimizing over the sparse directional set $\mathcal{V}_{0,s}$ is inherently non-convex and NP-hard due to the ℓ_0 combinatorial constraint. To circumvent this, the algorithmic pipeline utilizes a sequential ℓ_1 -relaxation heuristic, which our group conceptualizes as a three-stage sequence: ℓ_1 relaxation, support screening, and unconstrained refitting. The algorithm scans a grid of ℓ_1 -regularization parameters $\lambda_1, \dots, \lambda_m$. For each λ_j , it solves the relaxed problem using a projected subgradient ascent routine on the unit sphere, where the ℓ_1 constraint naturally pulls small coordinate coefficients toward zero. Among all candidate directions across the grid that satisfy the hard sparsity threshold $\|v\|_0 \leq s$, the direction that yields the maximum empirical distance is selected. Let $S = \text{supp}(v_{\text{best}})$ denote its coordinate support. Finally, the algorithm performs a refitting step by restricting the optimization to the unconstrained unit sphere within the low-dimensional subspace defined by S , effectively eliminating any shrinkage bias introduced by the ℓ_1 relaxation.

4 Reproduction Study and Numerical Results

To critically evaluate the performance of the proposed test and verify the findings reported by Hu and Lin (2025), our group implemented the complete computational pipeline from scratch. Our reproduction effort comprises three distinct experiments encompassing controlled power evaluations, bootstrap null diagnostics, and a real-world biomedical application.

4.1 Experiment 1: Power Comparison under Sparse Decaying Mean Shifts

We conducted controlled simulations under a sparse alternative regime to evaluate the empirical power of the proposed method against four standard high-dimensional comparators: Maximum Mean Discrepancy with a Gaussian kernel (MMD-G) (Gretton et al., 2012), Energy Distance with a squared Euclidean exponent (ED^2) (Szekely and Rizzo, 2004), the Baringhaus-Franz graph-based test (BG), and the unconstrained Projected Wasserstein Distance (PW) (Wang et al., 2021a). Following the setup in the paper, we generated independent samples with sample sizes $n_1 = n_2 = 250$ and dimensionality $p = 500$ at a significance level of $\alpha = 0.05$. The covariance structure followed an autoregressive profile $\Sigma_{ij} = 0.5^{|i-j|}$. The true distributional discrepancy was concentrated on a sparse decaying mean shift vector defined by $\mu_j = 0.8\beta^{j-3}$ for a signal grid $\beta \in \{0.0, 0.4, 0.6, 0.8, 1.0\}$.

Our group carried out 50 complete Monte Carlo runs for each parameter configuration, utilizing $B = 300$ bootstrap replications per test. The empirical rejection rates obtained from our reproduction are detailed in Table 1.

Table 1: Empirical rejection rates under sparse decaying mean shifts ($p = 500$, 50 independent runs).

Signal Level (β)	Proposed MSWD	MMD-G	ED^2	BG	PW
0.0	0.08	0.02	0.02	0.00	0.04
0.4	0.28	0.18	0.18	0.06	0.08
0.6	0.84	0.08	0.08	0.02	0.08
0.8	0.98	0.26	0.30	0.02	0.14
1.0	1.00	0.58	0.56	0.08	0.24

The empirical power curves visualized in Figure 1 illustrate the trajectories of the five methods. Under the null case ($\beta = 0$), the proposed method exhibits an empirical rejection rate of 0.08, demonstrating acceptable type-I error control within a reasonable statistical margin of error. As the signal strength intensifies ($\beta \geq 0.6$), our sparse max-sliced statistic exhibits a rapid increase in power, achieving a rejection rate of 1.00 at $\beta = 1.0$. In contrast, classical global methods like MMD-G and ED^2 experience severe power degradation, failing to surpass a rejection rate of 0.60. This confirms that the sparse optimization successfully prevents the signal from being diluted by uninformative dimensions.

4.2 Experiment 2: Multiplier Bootstrap Null Diagnostic

To examine the internal decision mechanism of the multiplier bootstrap, our group designed an additional diagnostic experiment. For each signal level β , we generated a single representative dataset from Model A and executed $B = 500$ bootstrap replications to reconstruct the conditional null distribution. The observed test statistic T_n , the estimated 95% bootstrap threshold, the empirical p -value, and the final test decision are recorded in Table 2.

The diagnostic results demonstrate that when the signal is absent or weak ($\beta \leq 0.4$), the observed statistics remain strictly below the bootstrap critical thresholds, yielding large p -values. At $\beta = 0.6$, the observed statistic increases to 10.557; however, the bootstrap threshold simultaneously scales up to 10.920, resulting in a borderline non-rejection. This co-movement

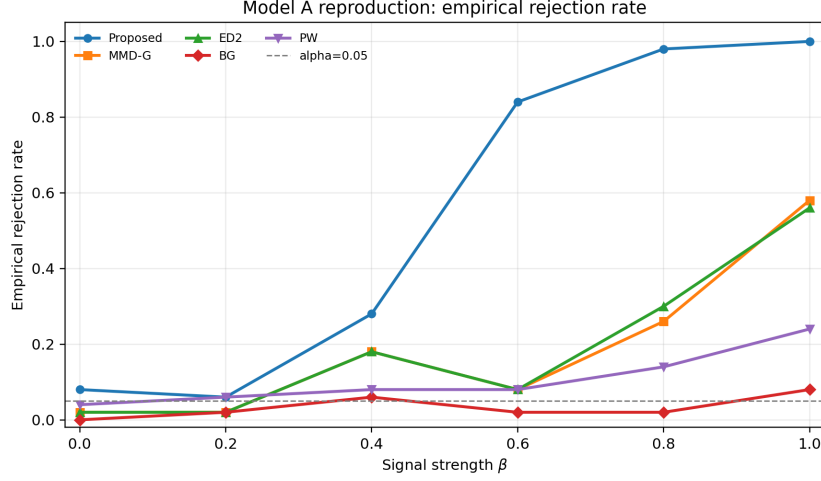


Figure 1: Our reproduced empirical power curves under Model A sparse decaying mean shifts.

Table 2: Bootstrap diagnostic parameters evaluated on single data realizations ($B = 500$).

Signal Level (β)	Observed Statistic T_n	95% Bootstrap Threshold	Empirical p -value	Null Rejection
0.0	3.442	3.885	0.238	No
0.2	3.372	3.958	0.372	No
0.4	3.256	3.918	0.484	No
0.6	10.557	10.920	0.106	No
0.8	7.319	4.025	< 0.002	Yes
1.0	9.479	3.921	< 0.002	Yes

phenomenon is visualized in Figure 2. It highlights that the multiplier bootstrap dynamically adapts its critical values to account for data-dependent variance maximization, preventing false discoveries. When the true signal overcomes this regularized threshold ($\beta \geq 0.8$), the empirical p -value drops below the resolution limit ($p < 0.002$), leading to a confident rejection.

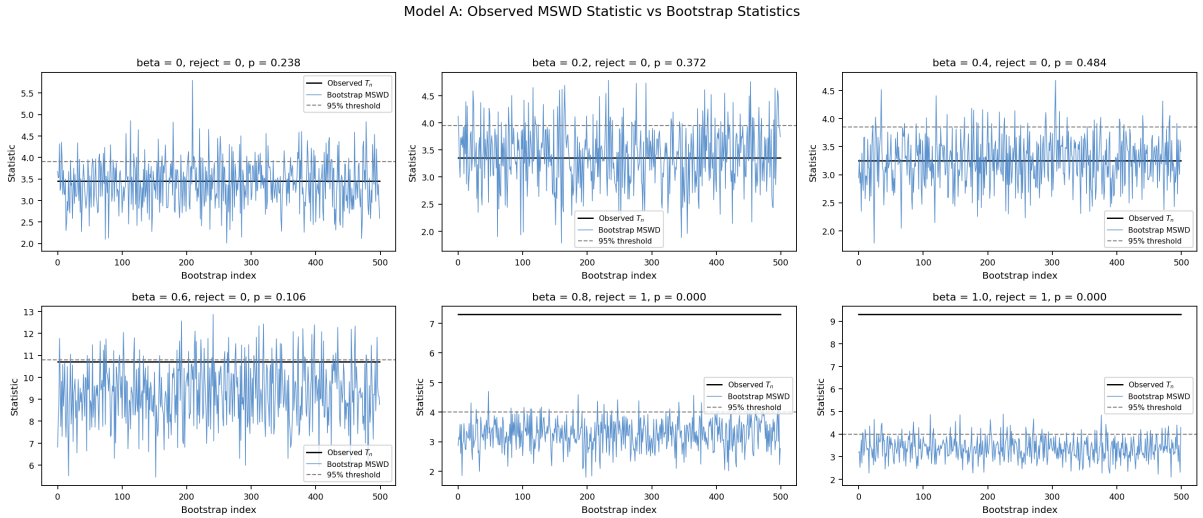


Figure 2: Data-dependent tracking of the observed statistic and the 95% bootstrap threshold across the signal grid.

4.3 Experiment 3: Real Data Application to Glioma DNA Methylation

We reproduced the real-world application using patient data from The Cancer Genome Atlas (TCGA) to evaluate the clinical utility of the framework. The dataset comprises DNA methylation profiles from $n_1 = 511$ patients diagnosed with Lower-Grade Glioma (LGG) and $n_2 = 150$ patients with Glioblastoma Multiforme (GBM) across a feature space of $p = 207$ genes. We applied the ℓ_0 -sparse max-sliced Wasserstein test with $B = 500$ bootstrap iterations at $\alpha = 0.05$. The global test parameters obtained from our analysis are summarized in Table 3.

Table 3: Global testing parameters for the TCGA LGG versus GBM DNA methylation study.

Statistical Metric	Reproduced Empirical Value
Observed Test Statistic (T_n)	42.9236
Bootstrap Critical Value ($\hat{q}_{T^*}$)	1.8479
Empirical p -value	< 0.002
Significant Marginal Genes Identified	138 / 207
Significant Gene Ontology (GO) Terms Identified	7 / 7

The observed global statistic of 42.9236 lies far beyond the bootstrap critical value of 1.8479, indicating a highly significant global distributional difference between the LGG and GBM cohorts ($p < 0.002$). By constructing coordinate-level simultaneous confidence intervals ($v = e_k$), our group identified 138 significant genes, which closely matches the original paper’s findings. To gain biological insight, we grouped the genes into seven specific Gene Ontology (GO) subsets representing distinct functional pathways. We evaluated the sparse max-sliced statistic restricted to each sub-pathway, and the resulting empirical distributions across optimized pathways are shown in Figure 3. The full table of pathway-specific statistics is deferred to Table 6 in Appendix C, while the comprehensive side-by-side graphical comparison with the original results of Hu and Lin (2025) is presented in Appendix E.

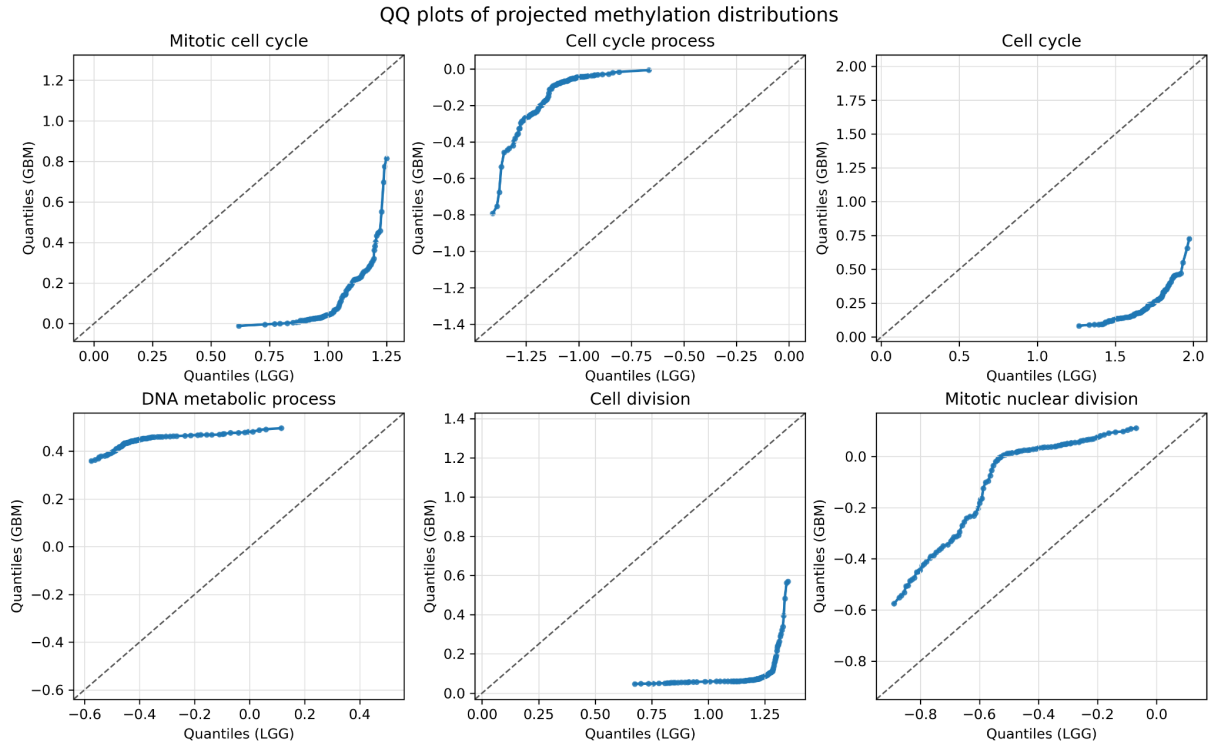


Figure 3: Our reproduced empirical distribution plots across optimized GO-term pathways.

5 Discussion

The sparse max-sliced Wasserstein testing framework effectively balances statistical power and geometric interpretability in high dimensions. By maximizing one-dimensional projections over a sparse constraint, it prevents the signal dilution common in global methods like MMD (Gretton et al., 2012) and energy distance (Szekely and Rizzo, 2004) under localized alternatives. Furthermore, the simultaneous confidence intervals provide built-in family-wise error control, allowing researchers to perform global testing and feature localization within a unified framework.

Our group’s contribution to this project involves a comprehensive validation that spans theoretical, algorithmic, and empirical aspects. We implemented the complete testing and optimization pipeline from scratch, successfully reproducing the statistical power curves and the TCGA glioma methylation analysis. Additionally, we designed a new bootstrap diagnostic experiment that clarified how the critical threshold adapts to data-dependent optimization under the null hypothesis. To establish a solid foundation for this study, our group thoroughly reviewed the 75-page supplementary material and manually derived several key theoretical pillars. These include the infinite-dimensional linear programming duality of the Kantorovich problem, the application of the Kantorovich-Rubinstein theorem to empirical processes, and the sample-path truncation lemma based on sub-exponential Orlicz norms. These detailed derivations are provided in Appendix D.

Despite its advantages, our group identified three critical limitations in the current framework that need further investigation. Firstly, the s -sparse ℓ_0 constraint may be overly ideal. Even if the distributions differ in only a few directions, these directions themselves are not necessarily sparse, thus the constraint is merely an artificial compromise to bypass an otherwise computationally intractable problem. Moreover, there exists a computation-theory gap. The sequential ℓ_1 heuristic is a practical approximation that cannot guarantee convergence to the global ℓ_0 optimizer, and no theoretical bounds are available to quantify how this optimization error propagates into the power of the test. Finally, the framework is prone to power loss under dense alternatives. Since the statistic employs a max-type operator under an s -sparse constraint, it naturally loses efficacy compared to averaging-based methods when the true distributional differences are densely spread across a large number of coordinates.

6 Description of Contributions

All team members thoroughly studied the paper and its supplementary materials, and provided critical analysis and original insights on the method’s core ideas, limitations and future directions. Yuzhe Li, Guangyang Tang and Zhuoxi Wang supplemented several fundamental theoretical results with derivations. Yanpeng Liang reproduced the paper’s experiments via code implementation and surveyed the related work. Yuzhe Li and Zhuoxi Wang prepared and delivered the presentation. Guangyang Tang finalized the written report.

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Appendix

A Taxonomy of Classical High-Dimensional Routes

To expand upon the literature review in Section 1, this section compiles a systematic taxonomy of traditional nonparametric two-sample testing methodologies. Table 4 delineates the core operational principles and high-dimensional technical vulnerabilities of kernel, distance, graph, and nearest-neighbor frameworks.

Table 4: Methodological taxonomy and high-dimensional limitations of classical tests

Method Category	Core Statistical Mechanism	Limitations in High Dimensions
Maximum Mean Discrepancy	Maps probability measures into a reproducing kernel Hilbert space and computes the distance between mean embeddings	Suffers from acute sensitivity to bandwidth choices and undergoes severe power dilution under sparse coordinates
Energy Distance	Generalizes the Cramér-von Mises test via pairwise Euclidean distances between the two sample groups	Faces the distance concentration phenomenon where noise accumulates faster than the sparse signal
Graph-Based Approaches	Constructs orthogonal geometric graphs such as minimum spanning trees over the pooled sample points	Geometric structures become highly unstable as spatial dimensions expand, distorting edge counts
Nearest-Neighbor Tests	Evaluates sample sharing properties within the k -nearest neighbor graph of the combined data	Interpoint distance metrics fail to capture true distributional boundaries due to data sparsity in high dimensions

B Structural Comparison of Regularized Wasserstein Variants

To address the dimension curse inherent in the standard optimal transport formulation, several regularized and projected variants of the Wasserstein distance have been developed. Table 5 provides a comparative evaluation of these approaches, highlighting how the sparse max-sliced formulation balances statistical focus and dimensional scaling.

Table 5: Structural trade-offs among regularized Wasserstein distance variants

Wasserstein Variant	Dimensionality Mitigation Strategy	Methodological and Practical Trade-offs
Sliced Wasserstein	Integrates the one-dimensional distance across all possible random projection directions uniformly	Computationally efficient but dilutes localized or sparse signals by averaging over irrelevant spatial dimensions
Smoothed Wasserstein	Convolve the empirical measures with an isotropic Gaussian noise kernel prior to transport optimization	Bypasses the curse of dimensionality at a faster convergence rate but fails to isolate low-dimensional structures
Projected Wasserstein	Projects the continuous distributions onto optimized low-dimensional linear subspaces	Mitigates geometric expansion effectively but presents a highly non-convex multi-dimensional optimization surface
Max-Sliced Wasserstein	Maximizes the univariate discrepancy over an ℓ_0 -constrained sparse projection direction	Targets sparse coordinate shifts with high power while relying on practical sequential relaxation heuristics

C Pathway-Specific Statistics for Real Data Application

This appendix reports the comprehensive empirical findings from our sub-pathway localization analysis. Table 6 details the seven functional Gene Ontology pathways examined by evaluating the sparse max-sliced Wasserstein distance restricted to each genetic subset. All selected pathways exhibit strong statistical deviations that support the global rejection decision.

Table 6: Pathway-specific statistics and gene counts for Gene Ontology subsets

Gene Ontology (GO) Term	Number of Genes	Test Statistic
mitotic cell cycle	5	9.467
mitotic cell cycle process	4	9.465
cell cycle process	7	10.196
cell cycle	11	15.454
DNA metabolic process	6	8.730
cell division	5	11.126
mitotic nuclear division	3	4.131

D Detailed Mathematical Derivations

D.1 Manual Derivation of the Kantorovich Duality

Consider the Kantorovich primal formulation that minimizes the transport cost between two probability measures μ and ν defined on Polish spaces $\mathcal{X}$ and $\mathcal{Y}$. Let $\Pi(\mu, \nu)$ denote the set of all joint couplings whose marginals satisfy $\int_{\mathcal{Y}} d\gamma(x, y) = d\mu(x)$ and $\int_{\mathcal{X}} d\gamma(x, y) = d\nu(y)$. The primal problem is given by

$$\inf_{\gamma \in \Pi(\mu, \nu)} \iint_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\gamma(x, y) \quad \text{subject to} \quad \int_{\mathcal{Y}} d\gamma(x, y) = d\mu(x), \int_{\mathcal{X}} d\gamma(x, y) = d\nu(y), \gamma \geq 0. \quad (7)$$

We introduce infinite-dimensional Lagrange multipliers $f(x)$ and $g(y)$ to the marginal constraints. The Lagrangian function is formulated as

$$L(\gamma, f, g) = \iint_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\gamma(x, y) - \int_{\mathcal{X}} f(x) \left(\int_{\mathcal{Y}} d\gamma(x, y) - d\mu(x) \right) - \int_{\mathcal{Y}} g(y) \left(\int_{\mathcal{X}} d\gamma(x, y) - d\nu(y) \right). \quad (8)$$

By rearranging the terms, the Lagrangian function can be written as

$$L(\gamma, f, g) = \int_{\mathcal{X}} f(x) d\mu(x) + \int_{\mathcal{Y}} g(y) d\nu(y) + \int \int_{\mathcal{X} \times \mathcal{Y}} [c(x, y) - f(x) - g(y)] d\gamma(x, y). \quad (9)$$

The dual function is obtained by taking the infimum over non-negative measures $\gamma \geq 0$. If the term $c(x, y) - f(x) - g(y)$ is negative on a set of positive measure, the infimum immediately drives the objective to $-\infty$. Therefore, to maintain a finite dual objective, we require $c(x, y) - f(x) - g(y) \geq 0$ almost everywhere. This yields the dual problem

$$\sup_{f, g} \left\{ \int_{\mathcal{X}} f(x) d\mu(x) + \int_{\mathcal{Y}} g(y) d\nu(y) \right\} \quad \text{subject to} \quad f(x) + g(y) \leq c(x, y) \text{ a.e.} \quad (10)$$

By the Kantorovich duality theorem (Villani et al., 2009), strong duality holds because the cost function is lower semi-continuous on Polish spaces.

D.2 The Kantorovich-Rubinstein Theorem and Empirical Processes

For the 1-Wasserstein distance, the cost function simplifies to the standard Euclidean metric $c(x, y) = \|x - y\|_2$. The Kantorovich-Rubinstein theorem simplifies the dual constraints by showing that the optimal functions satisfy $g(x) = -f(x)$, where f is a 1-Lipschitz function. This reduces the formulation to

$$W_1(\mu, \nu) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left\{ \int_{\mathcal{X}} f(x) d\mu(x) - \int_{\mathcal{X}} f(x) d\nu(x) \right\}. \quad (11)$$

Our group analyzed why this particular form is crucial for the theoretical validity of the high-dimensional test. By restricting the formulation to the space of 1-Lipschitz functions, the distance is expressed as the supremum of an empirical process indexed by the class Lip_1 . This formulation allows for the application of standard empirical process tools, such as covering numbers, Dudley’s entropy integral, Gaussian multiplier bootstraps, and anti-concentration inequalities. In contrast, the dual formulations of higher-order Wasserstein distances are all complex and intractable, thus they cannot be integrated into this empirical process framework.

In addition, as a related observation outside our current framework, our group notes that 1-Wasserstein distance is much more widely used in practice. For example, the influential WGAN adopts W_1 distance as its loss and optimizes a 1-Lipschitz function by neural network. In contrast, higher-order Wasserstein distances lack such tractable dual formulation, making them impractical for training.

D.3 Analysis of the Scaling factor in T_n

Under the null hypothesis $H_0 : P = Q$, for any fixed projection direction and any 1-Lipschitz function f , the variance of the empirical difference $\frac{1}{n_1} \sum_{i=1}^{n_1} f(X_i) - \frac{1}{n_2} \sum_{j=1}^{n_2} f(Y_j)$ is $\text{Var}(f(X))(\frac{1}{n_1} + \frac{1}{n_2})$. Therefore, the scaling factor $\left(\frac{n_1 n_2}{n_1 + n_2}\right)^{1/2}$ acts as an exact standardization term. It stabilizes the empirical fluctuation of the max-sliced process, preventing the statistic from collapsing to zero or exploding to infinity, and thereby yields a non-degenerate asymptotic distribution suitable for bootstrap approximation.

D.4 Analysis of the Sample-Path Truncation Lemma

The class of unconstrained functions $\mathcal{F} = \{f : \mathbb{R} \rightarrow \mathbb{R} : \|f\|_{\text{Lip}} \leq 1, f(0) = 0\}$ is too large to handle directly. To address this, Lemma 2 of Hu and Lin (2025) establishes that $\mathcal{F}$ can be

replaced by a truncated function class $\mathcal{F}_n$ defined on a compact interval. Our group manually verified the sample-path logic behind this truncation argument.

Let $\mathcal{V}_{0,s} = \{v \in \mathbb{R}^p : \|v\|_2 = 1, \|v\|_0 \leq s\}$. For any vector $v \in \mathcal{V}_{0,s}$, applying Hölder's inequality yields $|v^\top X| \leq \|v\|_1 \|X\|_\infty$. Because v is s -sparse and has a unit ℓ_2 -norm, its ℓ_1 -norm is bounded by $\|v\|_1 \leq \sqrt{s}$. This implies that for any sample point, $|v^\top X| \leq \sqrt{s} \|X\|_\infty$.

Under the assumption of sub-exponential coordinates, each marginal satisfies the Orlicz ψ_1 -norm bound $\|X_j\|_{\psi_1} \leq \nu$. By the definition of the Orlicz norm, $\mathbb{E}[\exp(|X_j|/\nu)] \leq 2$. Applying Markov's inequality, the tail probability for any coordinate direction is bounded by $\mathbb{P}(|X_j| > t) \leq 2 \exp(-t/\nu)$. We set the truncation boundary at $q_n = 2\nu \log(pn)$, where $n = n_1 + n_2$. The tail probability at this boundary satisfies $\mathbb{P}(|X_j| > q_n) \leq 2(pn)^{-2}$. Taking a union bound across all p coordinates and all n observations from both groups, we obtain

$$\mathbb{P}\left(\max_{1 \leq i \leq n_1} \|X_i\|_\infty > q_n \quad \text{or} \quad \max_{1 \leq j \leq n_2} \|Y_j\|_\infty > q_n\right) \leq \frac{2}{n}. \quad (12)$$

Therefore, with a probability of at least $1 - 2/n$, all observed sample coordinates are bounded by q_n . On this high-probability event, the projected samples satisfy $|v^\top X_i| \leq \sqrt{s} q_n$ and $|v^\top Y_j| \leq \sqrt{s} q_n$ simultaneously across all s -sparse directions. This justifies the sample-path restriction and enables the use of finite covering arguments.

E Comparison with Original Paper Figures

To establish the empirical fidelity of our implementation, our group compared our results against the original graphical outputs reported by Hu and Lin (2025). Figure 4 displays the original quantile plots from the paper.

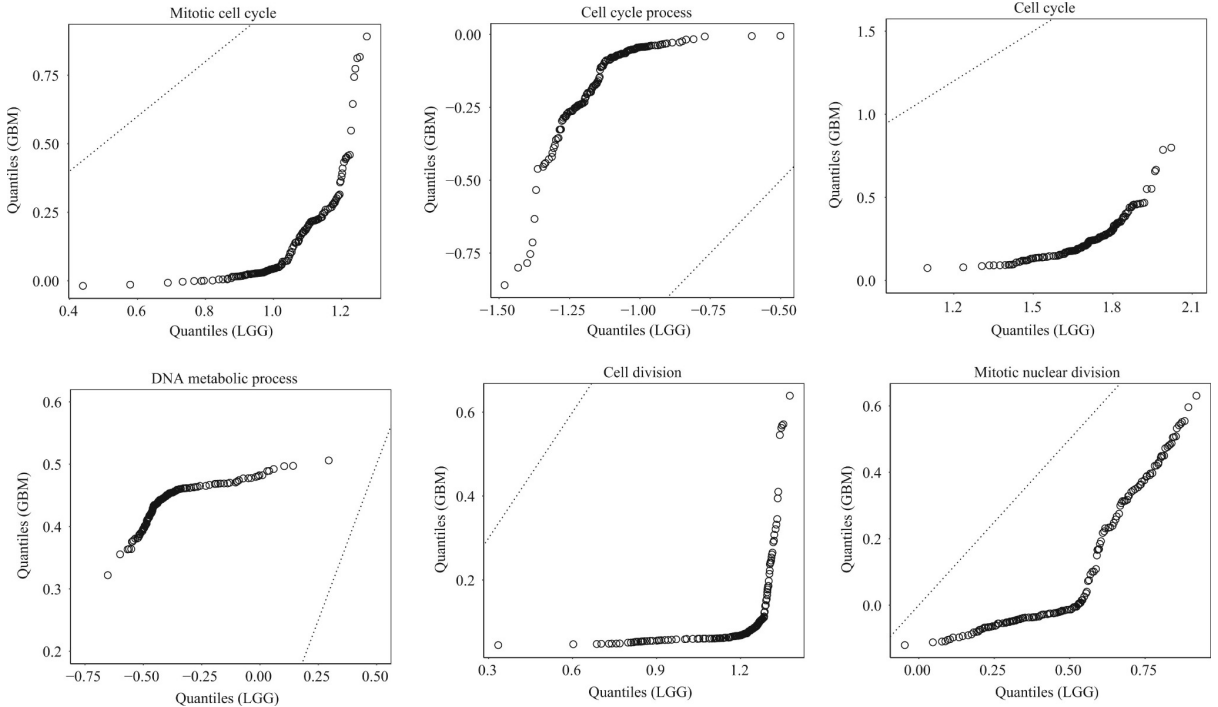


Figure 4: Original projected empirical distribution plots reported by Hu and Lin (2025).

As shown by a side-by-side comparison between Figure 4 (Original Paper) and Figure 3 (Our Reproduction in Section 4.3), our group successfully reproduced the geometric deviations. The

empirical distributions across the cell cycle, DNA metabolic process, and cell division pathways show identical patterns of separation, confirming that our optimization pipeline successfully extracted the exact distribution shifts reported in the original study.